

Degrees of Freedom (df) in Statistics

1. What is Degrees of Freedom?

The Degrees of Freedom (df) refers to the number of independent values that can vary in a statistical calculation. It is a mathematical constraint on a dataset, ensuring that statistical tests work properly.

- General Formula:

$$df = \text{Total number of observations} - \text{Number of estimated parameters}$$

- It determines the shape of the distribution used in tests like the t-distribution, chi-square, and F-distribution.
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2. Degrees of Freedom in Different Statistical Tests

A. Degrees of Freedom in a T-Test

1. One-Sample T-Test

$$df = n - 1$$

- Example: If you have a sample of 20 students and estimate the mean, you have $df = 20 - 1 = 19$.

2. Two-Sample (Independent) T-Test

$$df = (n_1 - 1) + (n_2 - 1)$$

- Example: If group 1 has 15 people and group 2 has 12, then:

$$df = (15 - 1) + (12 - 1) = 14 + 11 = 25$$

3. Paired T-Test

$$df = n - 1$$

- Used for before-and-after studies (same subjects tested twice).
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B. Degrees of Freedom in Chi-Square Test

1. Chi-Square Goodness of Fit Test

$$df = k - 1$$

- Where k is the number of categories.
- Example: If there are 4 categories (A, B, C, D), then:

$$df = 4 - 1 = 3$$

2. Chi-Square Test for Independence

$$df = (r - 1) \times (c - 1)$$

- Where r = number of rows, c = number of columns.
- Example: A survey records the relationship between Gender (Male/Female) and Shopping Preference (Online/In-Store):

	Online	In-Store	Total
Male	50	30	80
Female	70	50	120
Total	120	80	200

- Rows (r) = 2 (Male, Female)
 - Columns (c) = 2 (Online, In-Store)
 - $df = (2 - 1) \times (2 - 1) = 1 \times 1 = 1$
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C. Degrees of Freedom in ANOVA (Analysis of Variance)

1. One-Way ANOVA

- Between Groups: $df = k - 1$
- Within Groups: $df = n - k$
- Total df: $df = n - 1$
- Example: Testing the effect of 3 different diets on weight loss in 30 participants:
 - df Between Groups = $3 - 1 = 2$
 - df Within Groups = $30 - 3 = 27$
 - df Total = $30 - 1 = 29$

2. Two-Way ANOVA

- Factor A df = $a - 1$
 - Factor B df = $b - 1$
 - Interaction df = $(a - 1)(b - 1)$
 - Error df = Total sample size - (Number of groups)
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D. Degrees of Freedom in Regression

1. Simple Linear Regression

- df for Regression: $df = 1$ (one independent variable)
- df for Residual/Error: $df = n - 2$
- df for Total: $df = n - 1$

2. Multiple Regression

- df for Regression: $df = p$ (number of predictors)
 - df for Residual/Error: $df = n - p - 1$
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3. Summary Table

Test Type	Formula for Degrees of Freedom	Example Calculation
One-Sample T-Test	$df = n - 1$	$df = 25 - 1 = 24$
Two-Sample T-Test	$df = (n_1 - 1) + (n_2 - 1)$	$df = (15 - 1) + (12 - 1) = 25$
Chi-Square Goodness of Fit	$df = k - 1$	$df = 5 - 1 = 4$
Chi-Square Test for Independence	$df = (r - 1) \times (c - 1)$	$df = (3 - 1) \times (2 - 1) = 2$
One-Way ANOVA	$df_{between} = k - 1,$ $df_{within} = n - k$	$df = 4 - 1 = 3, df = 20 - 4 = 16$
Regression (Simple)	$df_{regression} = 1, df_{residual} = n - 2$	$df = 30 - 2 = 28$

4. Why Are Degrees of Freedom Important?

- Affects test statistics (t-values, chi-square values).
- Determines the shape of distributions (t-distribution, F-distribution).
- Influences p-values → Higher degrees of freedom **reduce variability**.